HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F78PA

PROBABILITY AND STATISTICS A

Semester 1 2022/23

Duration: Two Hours

There are 6 questions in this paper, 85 marks in total.
Candidates must attempt all 6 questions.
To receive full credit you must show your work and explain your answers, i.e. a correct answer without explanation may yield no credit.

Approved electronic calculators may be used.

AMS formula sheet is provided at the start of this exam paper.
'New Cambridge Statistical Tables' are provided.

Some Absolutely Continuous Distributions

Name	Notation	p.d.f.	range	m.g.f.	mean	variance
Uniform	$U(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{b\theta}-e^{a\theta}}{(b-a)\theta}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu\theta+\frac{1}{2}\sigma^2\theta^2}$	μ	σ^2
Gamma	$\Gamma(t,\lambda)$	$\frac{\lambda^t}{\Gamma(t)}x^{t-1}e^{-\lambda x}$	$0<x<\infty$, $t,\lambda>0$	$\left(\frac{\lambda}{\lambda-\theta}\right)^t$	$\frac{t}{\lambda}$	$\frac{t}{\lambda^2}$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-\theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	$0<x<\infty$, $n>0$ integer	$(1-2\theta)^{-n/2}$	n	$2n$
Beta	$\beta(\alpha_1,\alpha_2)$	$\frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}$	$0<x<1$, $\alpha_1,\alpha_2>0$	not useful	$\frac{\alpha_1}{\alpha_1+\alpha_2}$	$\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$
Cauchy	–	$\frac{\lambda}{\pi(\lambda^2+(x-\mu)^2)}$	$-\infty<x<\infty$, $\lambda>0$	doesn't exist but char. fn. is $e^{i\mu\theta-\lambda\lvert\theta\rvert}$	no moments exist	
Multivariate Normal	$N(\mu,\Sigma)$	$\frac{\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}$	$x\in\mathbb{R}^d$	$e^{\mu^T\theta+\frac{1}{2}\theta^T\Sigma\theta}$	μ	covariance matrix Σ

Some Discrete Distributions
(All distributions below have integer-valued ranges)

Name	Notation	prob. fn.	range	m.g.f.	mean	variance
Binomial	$\text{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$0\le x\le n$ $0<p<1$	$[(1-p)+pe^\theta]^n$	np	$np(1-p)$
Poisson	$\text{Pois}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$0\le x<\infty$ $\lambda>0$ real	$e^{\lambda(e^\theta-1)}$	λ	λ
Hyper-geometric	$\text{Hgeo}(n,a,b)$	$\frac{\binom{a}{x}\binom{b}{n-x}}{\binom{a+b}{n}}$	$0\le x\le n$ $a,b>0$ integers	not useful	$\frac{na}{a+b}$	$\frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$
Negative binomial	$\text{Nbin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$k\le x<\infty$, $0<p<1$	$p^k[e^{-\theta}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$
Geometric	$\text{Geo}(p)$	$p(1-p)^{x-1}$	$1\le x<\infty$ $0<p<1$	$p[e^{-\theta}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$

Multinomial: multivariate extension of binomial

Prob. fn. is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_!!x_2!\ldots x_k!x_{k+1}!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k}p_{k+1}^{x_{k+1}},\quad 0\le x_i\le n,\ \ 0<p_i<1\ \ \forall i=1,2,\ldots,n+1$$

where $p_1+p_2+\cdots+p_{k+1}=1$ and $x_1+x_2+\cdots+x_{k+1}=n$.

The m.g.f. is given by

$$[p_1e^{\theta_1}+p_2e^{\theta_2}+\cdots+p_ke^{\theta_k}+p_{k+1}]^n.$$

Moments: $E(X_i)=np_i$, $\text{Var}(X_i)=np_i(1-p_i)$, $\text{cov}(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n+1$.

1. The hourly wages (in £) of a sample of 15 workers in a factory are as follows:

3.40	3.60	3.75	3.95	4.20	4.50	4.75	5.20
5.75	6.45	6.60	7.25	8.75	9.60	12.25	

Note that for this sample of data, $\sum x = 90.0$ and $\sum x^2 = 631.25$.

(a) Calculate the median and inter-quartile range of the hourly wages of the workers.

[3 marks]

(b) Draw a boxplot of the given data set and comment on the shape of the distribution.

[3 marks]

(c) Calculate the sample mean and sample standard deviation of hourly wages of the workers.

[3 marks]

(d) Some workers feel that they are being underpaid. After negotiation with the factory management, the workers are offered a choice of the following pay rises.

Scheme (I): an increase of £0.30 per hour;
Scheme (II): a 5% increase in hourly rates.

Use your answers from part (c) to calculate the mean and standard deviation of hourly wages of the 15 workers under schemes (I) and (II).

Explain why the management would not mind which scheme was implemented, but the workers might.

[4 marks]

[Total 13 marks]

[PLEASE TURN OVER]

2. A company has three factories producing a particular component, in Aberdeen, Belfast and Cardiff. 30% of the components are produced in Aberdeen, 30% in Belfast, and 40% in Cardiff. Of those components produced in Aberdeen, 1% are defective; of those produced in Belfast, 3% are defective; and of those produced in Cardiff, 2% are defective.

(a) A component is chose at random from those produced by the company. Calculate the probability that this component is defective.

[2 marks]

Given that a component, chosen at random, is found to be defective, calculate the probability that it was produced at the Aberdeen factory.

[2 marks]

(b) A sample of 20 components is taken from all those produced by the company. Denoting by X the number of defective components in the sample, then under reasonable assumptions, X is binomially distributed.

What assumption(s) must we make to justify our claim that X is binomially distributed?

[2 marks]

Calculate the probability that more than 2 components in the sample are defective.

[3 marks]

[Total 9 Marks]

[PLEASE TURN OVER]

3. (a) Suppose the random variable T is exponentially distributed with parameter $\lambda > 0$, so that T has probability density function

$$f_T(t) = \begin{cases} \lambda \mathrm{e}^{-\lambda t} & t \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Derive expressions for the (cumulative) distribution function $F_T(t)$ of T, and the expectation $\mathbf{E}(T)$. [5 marks]

(b) Suppose now that T is exponentially distributed with parameter $\lambda = 2$, and that $Y = 1 - \mathrm{e}^{-T}$.

i. Find the cumulative distribution function $F_Y(y)$ of Y. [6 marks]

ii. Show that the probability density function $f_Y(y)$ of Y is given (within the range of non-zero density) by

$$f_Y(y) = 2(1-y),$$

and specify the range of non-zero density. [3 marks]

iii. Find $\mathbf{E}[Y]$ and $\mathbf{Var}[Y]$. [3 marks]

[Total 17 Marks]

[PLEASE TURN OVER]

4. Suppose the cumulative distribution function $F_X(x)$ of a random variable X is given as follows:

$$F_X(x) = \begin{cases} 0 & x < 0, \\ \frac{x^2}{16} & 0 \leq x < 2, \\ \frac{1}{4} & 2 \leq x < 4, \\ \frac{x-3}{4} & 4 \leq x < 7, \\ 1 & x \geq 7. \end{cases}$$

(a) Sketch the graph of $F_X(x)$ and state the properties that it must satisfy in order to be a valid cumulative distribution function. [5 marks]

(b) Does the random variable X have a probability density function? Justify your answer. [3 marks]

(c) Determine:

i. $\Pr(1 < X \leq 5)$

ii. $\Pr(X = 4)$

iii. $\Pr(X > 3 \mid X \leq 5)$

[4 marks]

(d) Let $Y = 2X - 1$. Find $\mathbf{E}(Y)$. [3 marks]

[Total 15 Marks]

[PLEASE TURN OVER]

5. (a) Let X and Y be jointly continuous random variables, with joint probability density function $f_{X,Y}(x,y)$ given by

$$f_{X,Y}(x,y) = \begin{cases} cx+1 & x,y \geq 0,\ x+y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

i. Sketch the region where $f_{X,Y}(x,y) > 0$ and show that $c = 3$. [3 marks]

ii. Find the marginal probability density function $f_X(x)$ of X and show that for $0 \leq x < 1$, the conditional density of Y given $X = x$ is given by

$$f_{Y|X}(y \mid x) = \begin{cases} \dfrac{1}{1-x} & 0 \leq y \leq 1-x, \\ 0 & \text{otherwise.} \end{cases}$$

Identify the conditional distribution of Y given $X = x$ (that is, name the distribution and give any parameter values). [5 marks]

iii. For $0 \leq x < 1$, write down an expression for $\mathbf{E}(Y \mid X = x)$. Hence find $E(Y)$. [3 marks]

iv. Determine $\Pr(Y < 2X^2)$. [4 marks]

(b) Suppose U and V are independent random variables. Show that $\mathbf{Cov}(U, UV) = \mathbf{Var}(U)\mathbf{E}(V)$. [3 marks]

[Total 18 Marks]

[PLEASE TURN OVER]

6. Suppose a fair six-sided die is rolled n times. For $i = 1, 2, \ldots, n$, denote by X_i the number rolled on the ith roll, so that X_i has probability mass function $p_X(x) = \frac{1}{6}$ for $x = 1, 2, 3, 4, 5, 6$, and you may assume that $X_1, X_2, \ldots, X_n$ are independent random variables. Let $S_n = \sum_{i=1}^{n} X_i$.

(a) Calculate $\mathbf{E}(X_i)$ and $\mathbf{Var}(X_i)$. [3 marks]

(b) With $n = 100$, use the Central Limit Theorem to approximate $\Pr(S_{100} \geq 380)$. [6 marks]

(c) With $n = 6$, apply Chebyshev's inequality to determine an upper bound for $\Pr(S_6 \geq 50)$. [4 marks]

[*Hint: Chebyshev's inequality states that for a random variable Y with $\mathbf{E}(Y) = \mu$ and $\mathbf{Var}(Y) = \sigma^2 < \infty$, for any $\epsilon > 0$,*

$$\Pr(|Y - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}.$$]

[Total 13 Marks]

[END OF PAPER]